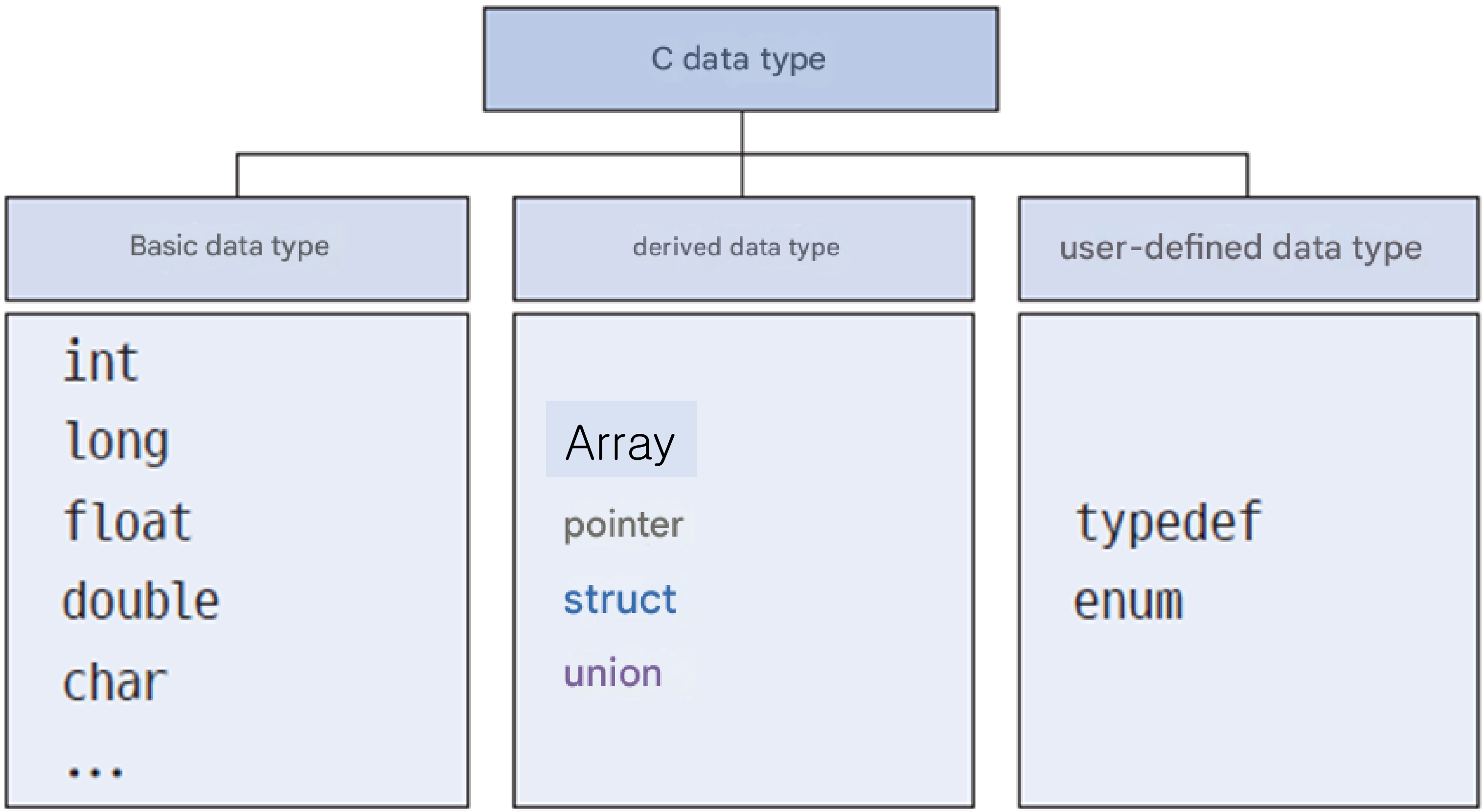
Ch.13 Structure

What you will learn in this chapter

- Structure concept, definition, and initialization method
- Relationship between structures and pointers
- Unions and typedefs

Structures are an important tool for grouping different data together .

Classification of data types



The need for structures

- How to gather data about students into one place ?

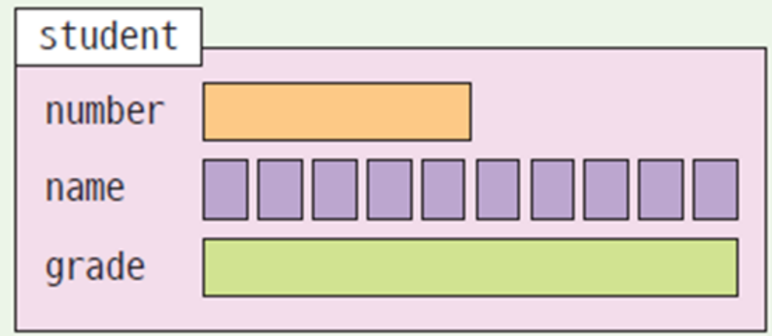
Student number: 20251234 (Number)
Name: “first last” (String)
Grade: 4.3 (Real number)
...

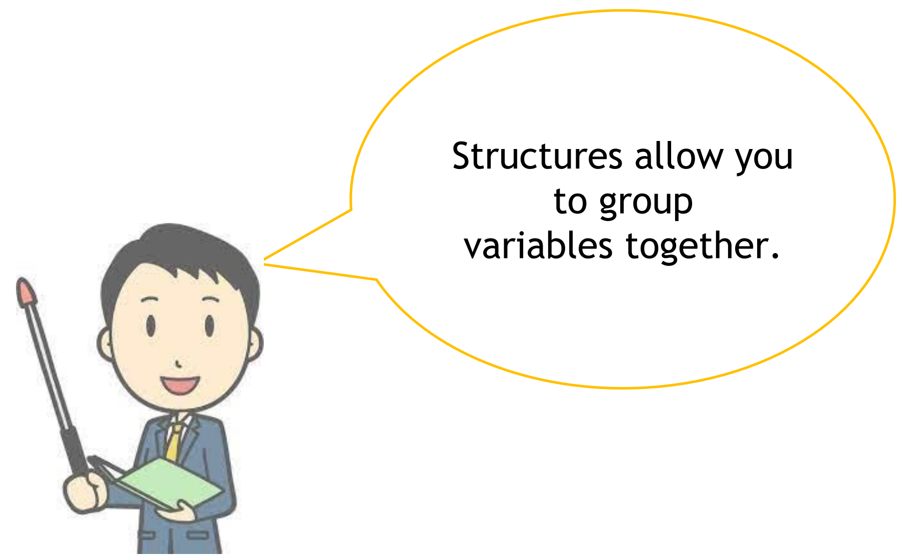
```c
int number;
char name[10];
double grade;
```

It can be expressed as individual variables like this, but can it be grouped ?

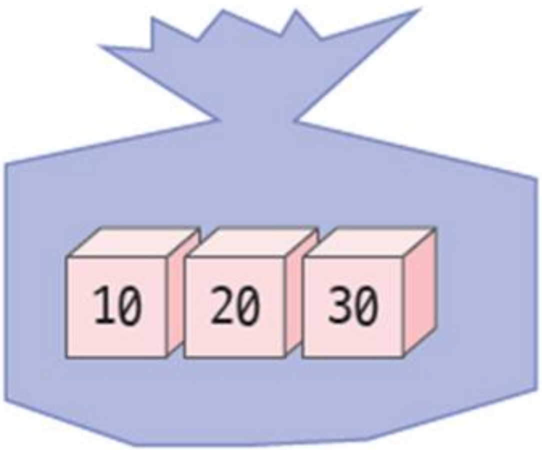
The need for structures

```
struct student {
        int number;        // student number
        char name[10];     // name
        double grade; //  unit
};
```

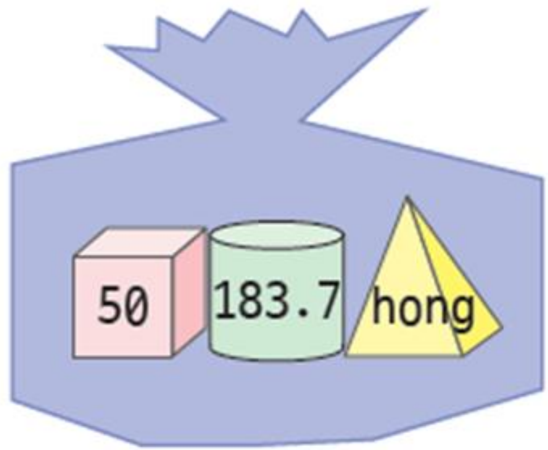




Arrays and Structures



Array

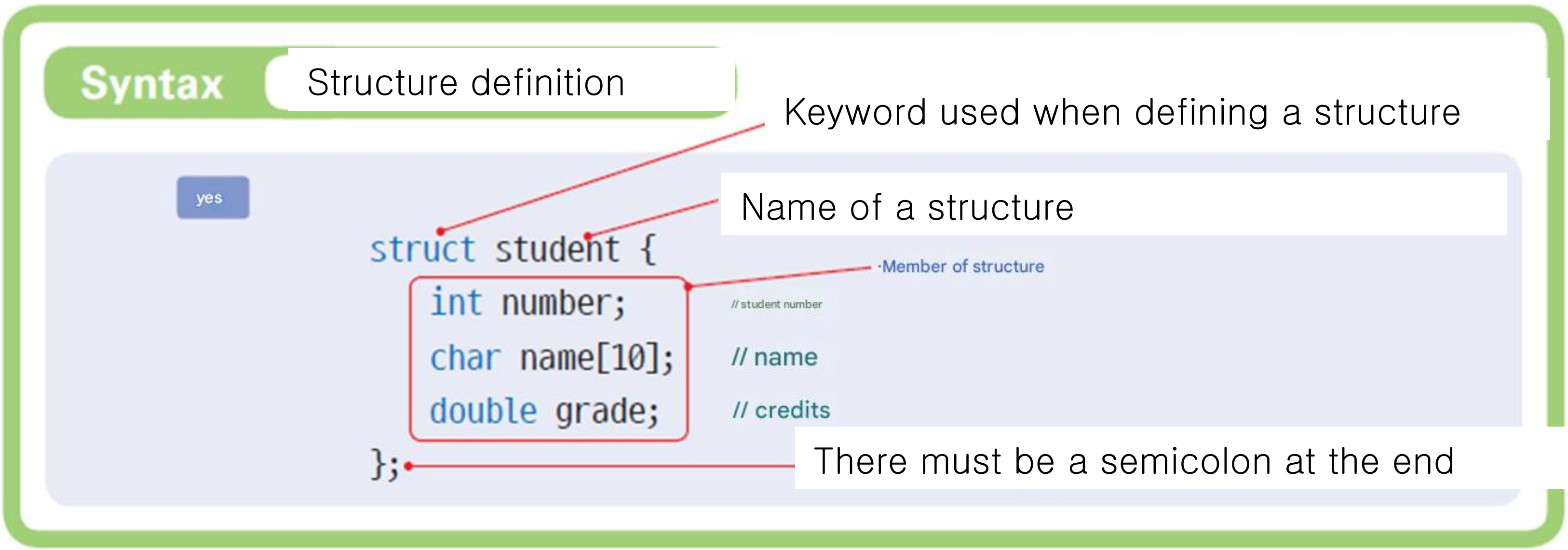


struct

Group variables of the same type

Group variables of different types

Structure definition



A *struct definition* creates the **blueprint** (the layout) of a structure type.

struct student p1;

This line is a **declaration of a variable** of type `struct student`.

- It declares a variable `p1`
- If `struct student` has already been defined, then this line also **allocates memory** for `p1`

Structure declaration

- *A structure definition is not a variable declaration.*

Defining a structure is
like a mold for making waffles.

To actually make waffles you need
to declare a structure variable.

Structure
ex) int

Structure variable
ex) int a

```
struct Person {
    int age;
};              // structure definition

struct Person p1;  // variable declaration + memory allocation
```

Example of a structure definition

```
// Screen coordinates made up of x
struct point {
        int x; // x coordinate
        int y; // y coordinate
};
```

```
// date
struct date {
        int month;
        int day;
        int year;
};
```

```
// Complex number
struct complex {
        double real; // real part
        double imag ; // imaginary part
};
```

```
// square
struct rect angle {
        int x , y ;
        int width;
        int height ;
};
```

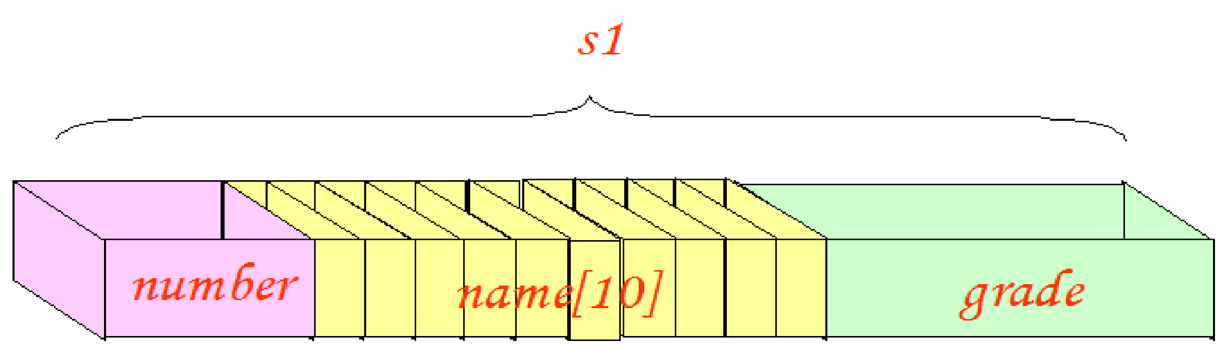
Declaring a structure variable

- Defining a structure and declaring a structure variable are different.

```
struct student {
        int number;
        char name[10];
        double grade;
};

int main(void){
        struct student s1;
        ...
}
```

Definition of Struct

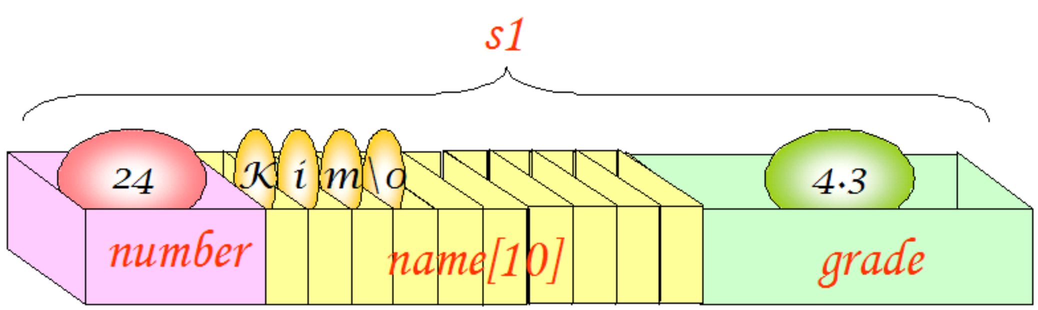
Declaration of Struct



Initializing a structure

- Initial values are listed using curly braces.

```
struct student {
        int number;
        char name[10];
        double grade;
};
struct student s1 = { 24, "Kim" , 4.3 };  // s1 is NOT a pointer,
                                           // it's variable of derived data type
```



Structure member reference

Syntax Accessing structure members

struct variable　　structure member

yes

```
s1.grade = 3.8;
```

.

. symbol is an operator used when referencing members in a structure

Example #1

Practice

```c
#include <stdio.h>
#include <string.h>
struct student {
        int number;
        char name[100];
        double grade ;
};

int main( void )
{
        struct student s;

        s.number = 20230001;
        strcpy (s.name, " Hong Gil-dong " );
        s.grade = 4.3 ;

        printf ( "Student number : %d\n" , s.number );
        printf ( "Name : %s\n" , s.name);
        printf ( “Grade : % .2f \n" , s.grade ) ;
        return 0;
}
```

Structure declaration

Declaring a structure variable

Structure member reference

Student number : 20230001
Name : Hong Gil-dong
GPA : 4.30

Example #2

Practice

```
struct student {
 int number;
 char name[10];
 double grade;
};

int main( void )
{
      struct student s;

      printf ( " Student number Enter : " );
      scanf ( "%d" , & s.number );

      printf ( " name Enter : " );
      scanf ( "%s" , s.name);

      printf ( " Credit Enter ( error ): " );
      scanf ( "%lf" , & s.grade );

      printf ( “ \ nStudent number : %d\n" , s.number );
      printf ( " Name : %s\n" , s.name);
      printf ( " Grade : % .2f \n" , s.grade );
      return 0;
}
```

Structure declaration

Declaring a structure variable

Passing the address of a structure member

```
Enter your student number : 20230001
Enter your name : Hong Gil-dong
Enter your grade ( error ): 4.3

Student number : 20230001
Name : Hong Gil-dong
GPA : 4.30
```

Initialization method

```
#include < stdio.h >
// Represents a point in two-dimensional space as a structure .
struct point {
	int x;
	int y;
};

int main( void )
{
	struct point p = { 1, 2 };              // ① Values are assigned in the order of the struct members.
	struct point q = { .y = 2, .x = 1 };   // ② Very readable, Order does not matter


	struct point r = p;                      // ③ structs can be copied using the = operator.
	r = ( struct point ) { 1, 2 };          // ④ Useful when you want to "reinitialize" after the variable is
	                                         //   declared.


	printf( "p=(%d, %d)\n" , p.x, p.y);
	printf( "q=(%d, %d) \n" , q.x, q.y);
	printf( "r=(%d, %d)\n" , r.x, r.y);
	return 0;
}
```

If you initialize only the first members, the rest become **zero** automatically:

Check points

1. Each variable declared within a structure is called _____.
2. The keyword used to declare a structure is ______.
3. Why are tags needed for structures, and what is the difference between using tags and not using them?
4. Are variables created just by declaring a structure?
5. What is the operator for referencing members of a structure?

Tag

```
struct Person {
   char name[20];
   int age;
};
```

A structure that has structures as members.

```c
struct date {
        int year;
        int month;
        int day;
};
```

```c
struct student {  // Structure declaration
        int number;
        char name[10];
        struct date d ; // structure within structure
        double grade;
};
```

```c
struct student s1; // Declare structure variable

s1.d.year = 1983;  // Member reference
s1.d.month = 03;
s1.d.day = 29;
```

Lab: Representing a rectangle as a point structure

- the point structure from the previous example to represent the coordinates of the vertices .

```
Enter the coordinates in the upper left corner : 10 10
Enter the coordinates in the upper right corner : 20 20
The area is 100 and the perimeter is 40 .
```

Example

```
#include < stdio.h >

struct point {
        int x;
        int y;
};

struct rect {
        struct point p1;
        struct point p2;
};

int main( void )
{
        struct rect r;
        int w, h, area, peri;
```

Example

```
printf( "Enter the coordinates of the upper left corner: " );
scanf( "%d %d" , &r.p1.x, &r.p1.y);

printf( "Enter the coordinates of the upper right corner: " );
scanf( "%d %d" , &r.p2.x, &r.p2.y);

w = r.p2.x - r.p1.x;
h = r.p2.x - r.p1.x;

area = w * h;
peri = 2 * w + 2 * h; (둘레)
printf( "Area is %d and perimeter is %d.\n" , area, peri);

 return 0;
}
```

Enter the coordinates in the upper left corner : 1 1
Enter the coordinates in the upper right corner : 6 6
The area is 25 and the perimeter is 20 .

Assignment and comparison of structure variables

- Assignment of variables of the same structure is possible, but comparison is not possible.

good way

```c
p2 = p1;
```

Not pointer,
It's a variable of derived data type

You can do this

```c
p2.x = p1.x;
p2.y = p1.y;
```

compilation error

```c
if( p1 == p2 )
{
        printf("p1 and p2 are equal.")
}
```

right way

```c
if( (p1.x == p2.x) && (p1.y == p2.y) )
{
        printf("p1 and p2 are equal.")
}
```

- Struct assignment is allowed because a struct is considered a single object.
  Structure assignment copies the entire memory block byte by byte.
- Array assignment is not allowed because arrays are treated as collections, not objects.

Example

- Assignment of variables of the same structure is possible, but comparison is not possible .

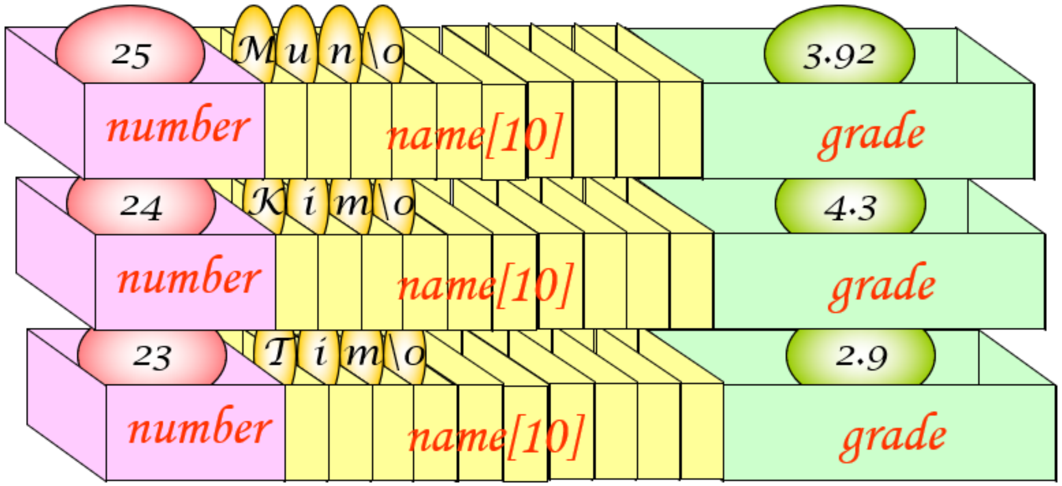
```
struct point {
      int x;
      int y;
};

int main( void )
{
      struct point p1 = {10, 20};
      struct point p2 = {30, 40};

      p2 = p1;    // Substitution possible

      if ( p1 == p2 )  // Compare -> Compile error !!
           printf ( " p1 and p2 It's the same ." )

      if ( (p1.x == p2.x) && (p1.y == p2.y) ) // Correct comparison
           printf ( " p1 and p2 It's the same ." )
}
```

Check points

1. What operations are allowed between variables of a structure ?
2. What is the difference between structure tags and structure variables ?
3. Can a structure be inserted as a structure member ?
4. Can a structure have an array as a member ?

Array of structures

- A collection of multiple structures



Array of structures

```
struct student {
	int number;
	char name[20];
	double grade;
};

int main( void )
{
	struct student list[100]; // Declare an array of structures


	list[2].number = 24;
	strcpy (list[2].name, " Hong Gil-dong " );
	list[2].grade = 4.3;
}
```

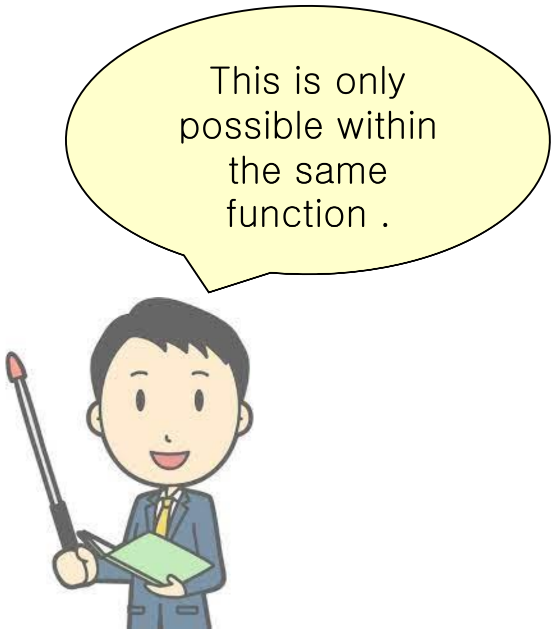
Array of structures initialization

```
struct student list[3] = {
      { 1, "Park", 3.42 },
      { 2, "Kim", 4.31 },
      { 3, "Lee", 2.98 }
};
```

Calculate the number of elements in an array of structures

- To automatically find the number of elements in an array of structure, Divide the total number of bytes in the entire array by the number of bytes in each element .

```
n = sizeof (list)/ sizeof (list[0]);
n = sizeof (list)/ sizeof (struct student);
```



```c
#include <stdio.h>
#define SIZE 3
struct student {
	int number;
	char name[20];
	double grade;
};

int main( void )
{
	struct student list[SIZE];
	int i;

	for ( i = 0; i < SIZE; i ++)
	{
	    printf ( " Student number Enter : " );
	    scanf ( "%d" , &list[i].number);
	    printf ( " name Enter : " );
	    scanf ( "%s" , list[i].name);
	    printf ( " Credit Enter ( error ): " );
	    scanf ( "%lf" , &list[i].grade);
	 }

	 for ( i = 0; i < SIZE; i ++)
	      printf ( " Student number : %d, Name : %s, Grade : %f\n" ,
			list[ i ].number, list[ i ].name, list[ i ].grade);
	 return 0;
}
```

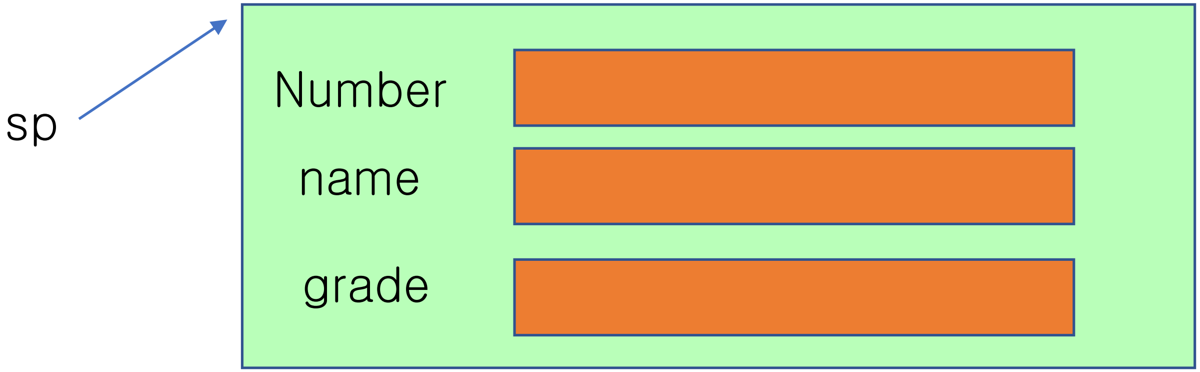
```
Enter your student number :
20190001
Enter your name : Hong Gil-dong
Enter your grade ( error ): 4.3
Enter your student number :
20190002
Enter your name : Kim Yu-shin
Enter your grade ( incorrect ): 3.92
Enter your student number :
20190003
Enter your name : Lee Seong-gye
Enter your grade ( incorrect ): 2.87
Name : Hong Gil-dong , Grade :
4.300000
Name : Kim Yu-shin , Grade :
3.920000
Name : Lee Seong-gye , Credit :
2.870000
```

Check points

1. Define an array of structures that can store information about five products. Products have number, name , and price as members .

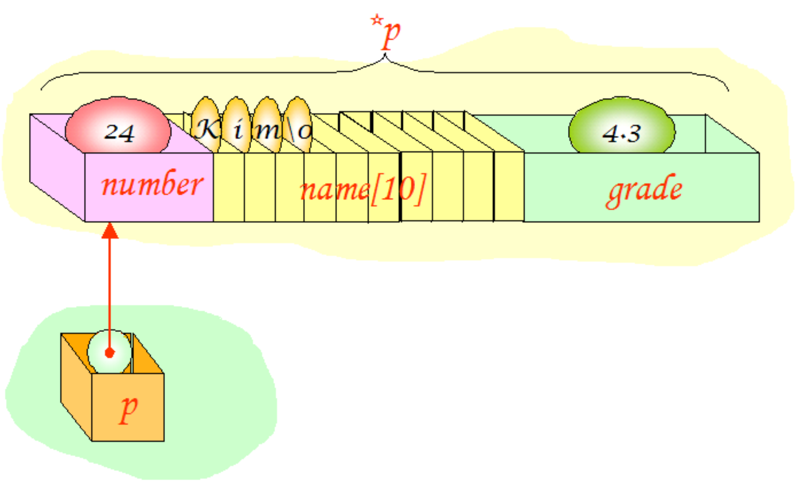
Structures and Pointers

1. Pointer to a structure
2. A structure that has pointers as members.



Pointer to a structure

```
struct student *p;

struct student s = { 24, “Kim”, 4.3 };
struct student *p;

p = &s;

printf("Student number =%d Name =%s Grade =%f \n" , s.number , s.name, s.grade );
printf("Student number =%d Name =%s Grade =%f \n" , (*p).number,(*p).name,(*p).grade);
```

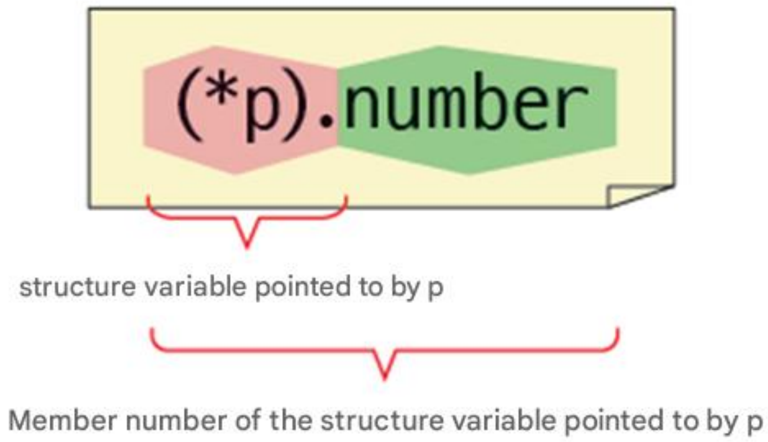


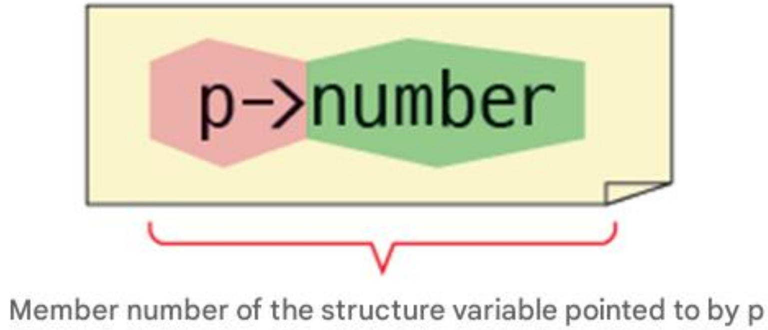
-> Operator

-> operator is used when referencing a structure member with a structure pointer.

```
struct student *p;

struct student s = { 24, “Kim”, 4.3 };
struct student *p;

p = &s;

printf("Student number =%d Name =%s Key =%f \n", p->number, p->name, p->grade);
```

-> Operator


(*p).number
structure variable pointed to by p
Member number of the structure variable pointed to by p


p->number
Member number of the structure variable pointed to by p

Example

Practice

```
// Referencing a structure through a pointer
#include <stdio.h>

struct student {
	int number;
	char name[20];
	double grade;
};

int main( void )
{
	struct student s = { 1, " Hong Gil-dong " , 4.3 };
	struct student * p;

	p = &s;

	printf("Student number=%d Name=%s Grade=%f\n" , s.number , s.name, s.grade );
	printf("Student number=%d Name=%s Grade=%f\n" , (*p).number, (*p).name, (*p).grade);
	printf("Student number=%d Name =%s Grade=%f\n" , p->number, p->name, p->grade);

	return 0;
}
```

```
Student number = 1 Name = Hong Gil-dong Grade = 4.300000
Student number = 1 Name = Hong Gil-dong Grade = 4.300000
Student number = 1 Name = Hong Gil-dong Grade = 4.300000
```

A structure that has pointers as members.

```
struct date {
	int month;
	int day;
	int year;
};

struct student {
	int number;
	char name[20];
	double grade;
	struct date *today ;
};
```

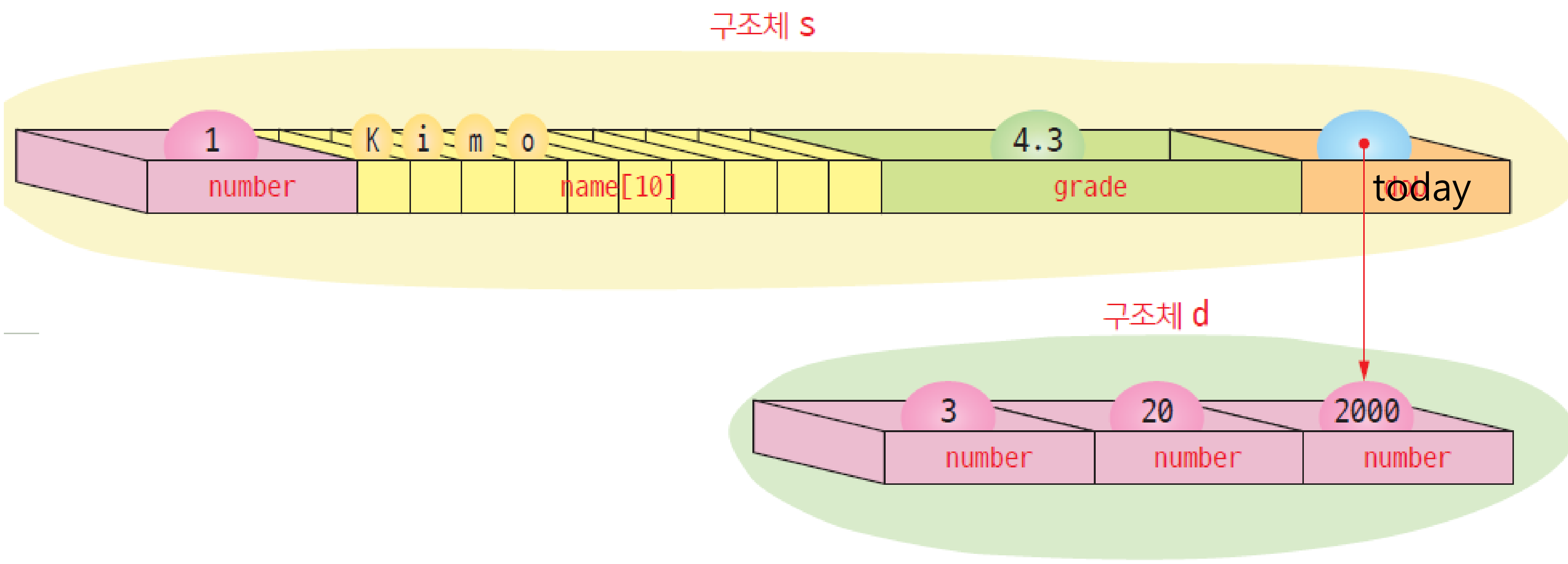
A structure that has pointers as members.

```
int main( void )
{
    struct date d = { 3, 20, 2000 };
    struct student s = { 1, "Kim" , 4.3 };

    s.today = &d;

    printf("Student number : %d\n" , s.number );
    printf("Name : %s\n" , s.name);
    printf("Grade : %f\n" , s.grade );
    printf("Birth: Year %d month %d day \n" ,
                                s.today ->year,
                                s.today ->month,
                                s.today ->day);

    return 0;
}
```

```
Student number : 1
Name : Kim
Credit : 4.300000
Birth : March 20 , 2000
```

Practice

A structure that has pointers as members.



Structures and functions

- When passing *a structure as an argument to a function*
  - ***A copy* of the structure** is passed to the function .
  - If the size of the structure is large, it takes more time and memory .

For structures, they are copied.

```c
int equal(struct student s1, struct student s2)
{
        if( s1.number == s2.number )
            return 1;
        else
            return 0;
}
```

```c
int main(void)
{
    struct student a = { 1, "lee", 3.8
};
    struct student b = { 2, "kim", 4.0 };
    if( equal(a, b) == 1 ){
        printf("Same student \n");
    }
    else {
        printf("Other students \n");
    }
}
```

Structures and functions

- When passing ***a pointer to a structure as an argument to a function***
  - It can save time and space .
  - There is a possibility that the original may be modified .

```
int equal(struct student *p1, struct student *p2)
{
        if( p1->number == p2->number )
            return 1;
        else
            return 0;
}
```

```
int main(void)
{
    struct student a = { 1, "lee", 3.8
};
    struct student b = { 2, "kim", 4.0 };
    if( equal(&a, &b) == 1 ){
        printf("Same student \n");
    }
    else {
        printf("Other students \n");
    }
}
```

Send a structure pointer.

`Access the structure through a pointer.

No changes via pointers

- If you only need to read the original and do not need to modify it, you can use the const keyword when defining the parameter as follows:

```c
int equal(const struct student *p1, const struct student *p2)
{
      if( p1->number == p2->number )
          return 1;
      else
          return 0;
}
```

Modifying the structure
through this pointer is prohibited.

Returning a structure

- A copy is returned .

```c
struct student  create()
{
      struct student s;
      s.number = 3;
      strcpy(s.name,"park");
      s.grade = 4.0;
      return s;
}
```

```c
int main(void)
{
      struct student a;
      a = create();

      return 0;
}
```

Structure s is copied into structure a.

Check points

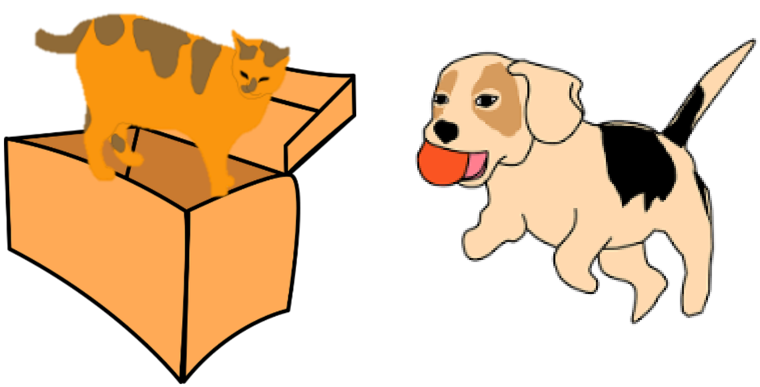
1. When passing a struct as an argument to a function, is the original passed or a copy passed?
2. When passing a pointer to a structure to a function , how can I avoid modiying the structure ?

Union

- union
  - Multiple variables share the same memory area
  - The way to declare and use a union is very similar to a structure.

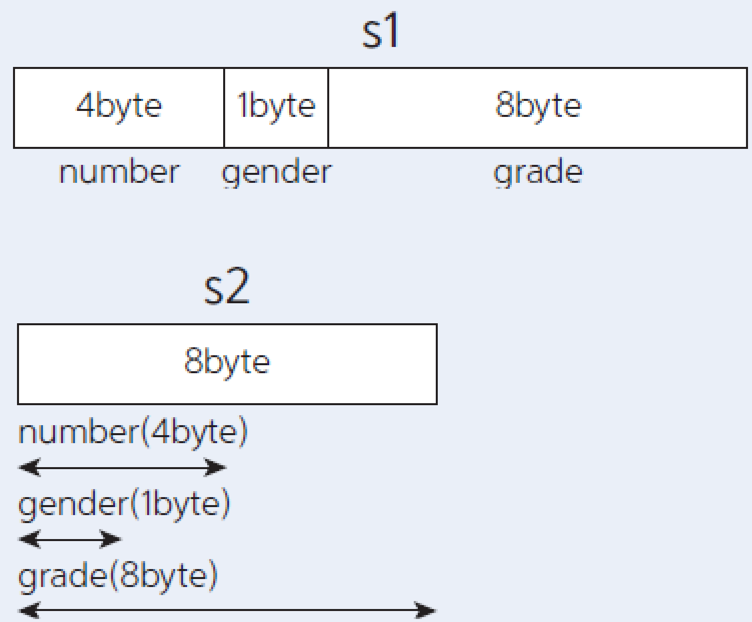
```
union example {
 char c;  // sharing the same space
 int i;     // sharing the same space
};
```

When Only One Data Type Is Used at a Time



Difference between structure and union

```
struct student
{
      int number;
      char gender;
      double grade;
} s1;

union student
{
      int number;
      char gender;
      double grade;
} s2;
```



Example

```
#include < stdio.h >

union example {
      int i ;
      char c;
};

int main( void )
{
      union example v;

      v.c = 'A' ;
      printf ( " v.c :%c v.i :% i \n" , v.c , v.i );

      v.i = 10000;
      printf ( " v.c :%c v.i :% i \n" , v.c , v.i );

}
```

Declaration of a union

Declaring a union variable .

Reference to char type .

Reference to type int .

```
vc:A vi :-858993599
vc :† vi:10000
```

Using type fields in unions

```
#include < stdio.h >
#include < string.h >
#define STU_NUMBER 1
#define REG_NUMBER 2

struct student {
      int type;
      union {
            int stu_number ;          // Student number
            char reg_number [15];  // resident registration number
      } id;
      char name[20];
};
```

Using type fields in unions

```
void print( struct student s)
{
	switch ( s.type )
	{
		case STU_NUMBER:
			printf ( " Student number %d\n" , s.id.stu_number );
			printf ( " Name : %s\n" , s.name);
			break;
		case REG_NUMBER:
			printf ( " Resident registration number : %s\n" , s.id.reg_number );
			printf ( " Name : %s\n" , s.name);
			break;
		default :
			printf ( " TypeError \n" );
			break;
	}
}
```

Using type fields in unions

```
int main( void )
{
	struct student s1, s2;

	s1.type = STU_NUMBER;
	s1.id.stu_number = 20190001;
	strcpy (s1.name, " Hong Gil-dong " );

	s2.type = REG_NUMBER;
	strcpy (s2.id.reg_number, "860101-1056076" );
	strcpy (s2.name, " Kim Cheol-su " );

	print(s1);
	print(s2);
}
```

```
Student number : 20190001
Name : Hong Gil-dong
Resident registration number : 860101-
1056076
Name : Kim Cheol-su
```

Check points

1. The keyword used to declare a union is _______ .
2. How is the size of memory allocated to a union determined ?

Definition & Declaration of enumeration

Syntax Enum definition

An enumeration is a data type that contains a collection of symbolic constants.

yes

```
enum days { SUN, MON, TUE, WED, THU, FRI, SAT };
```

When defining an enumeration

Keywords used

Name of the enumeration

Declaring enumeration variables

```
enum days today;
today = SUN; // OK!
```

variable today will be stored in memory as the integer value 0, because SUN maps to 0.

Compiler => #define SUN 0
#define MON 1
#define …. 2

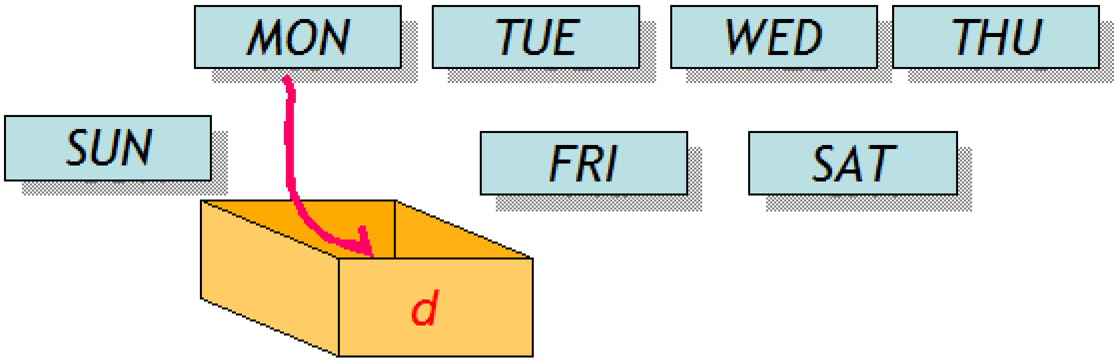
int today = SUN; // becomes: int today = 0;

Enumeration

- *An enumeration is* a data type: values lists in advance hat a variable can have.

enum days { SUN, MON, TUE, WED, THU, FRI, SAT};

(Example) A variable storing the day of the week can only have one of the following values :



Why we need enumerations

- You can write a program like this : Let's think about the problem

```
int today;
today = 0; // Sunday
today = 1; // Monday
…
```

- If you use enumerations,
  - It can reduce errors and improve readability.
  - The symbolic constant SUN is more preferable than 0, because its meaning is easier to understand.

Enumeration Initialization

```
enum days { SUN, MON, TUE, WED, THU, FRI, SAT }; // SUN=0, MON=1, ...
enum days { SUN=1, MON, TUE, WED, THU, FRI, SAT }; // SUN=1, MON=2, ...
enum days { SUN=7, MON=1, TUE, WED, THU, FRI, SAT=6 }; // SUN=7, MON=1,
...
```

If you do not specify a value,
it is assigned from 0.

Examples of enumerations

```
enum colors { white, red, blue, green, black };
enum boolean { false, true };
enum levels { low, medium, high };
enum car_types { sedan, suv , sports_car , van, pickup, convertible };
```

Example

```c
#include < stdio.h >
// Define an enum type for traffic light colors
enum TrafficLight {
    RED,    // 0
    YELLOW, // 1
    GREEN   // 2
};

int main(void) {
   enum TrafficLight light;

   light = RED;
   if (light == RED) {  printf("Stop!\n");   }

   light = GREEN;
   if (light == GREEN) {  printf("Go!\n");   }

   return 0;
}
```

Comparison of enumerations with other methods

Use of integers	Symbolic Constant	Enumeration

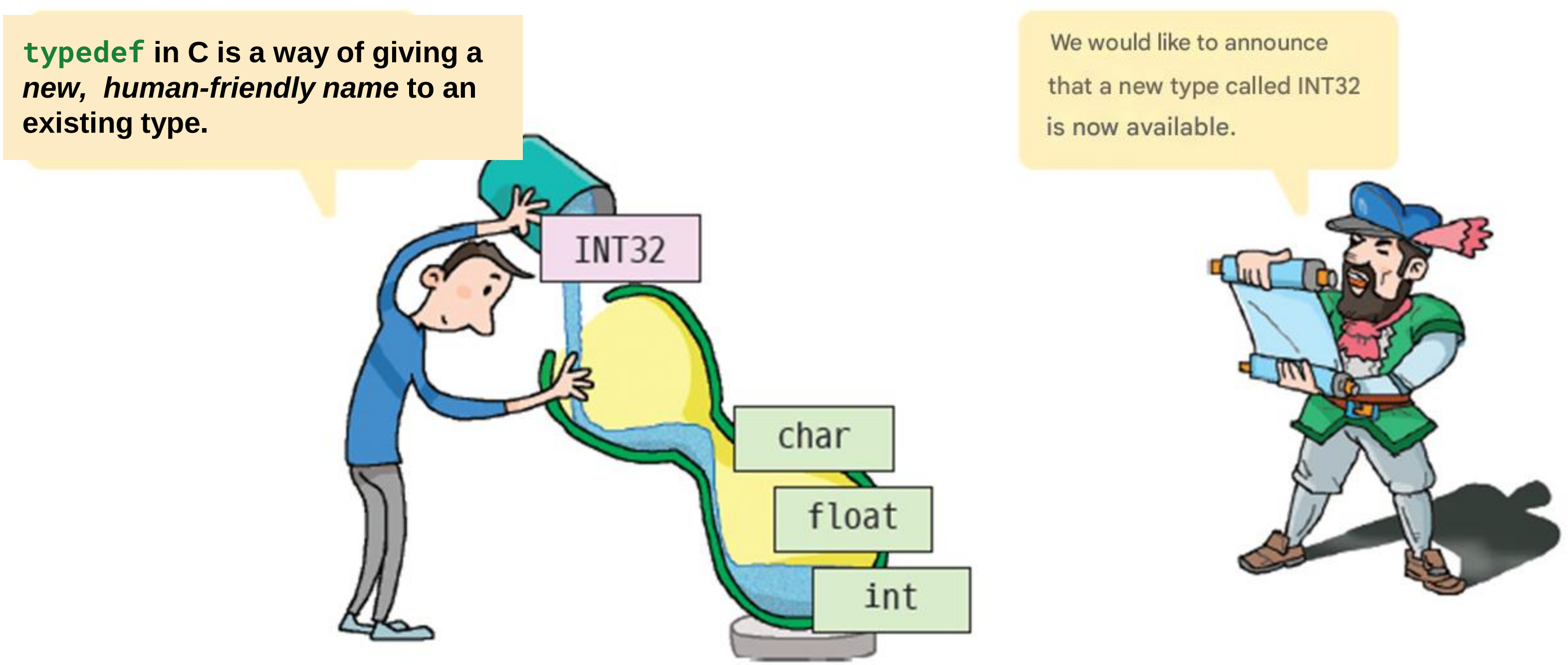
switch (code) {
  case 1:
      printf ("LCD TV\n");
      break ;
  case 2:
      printf (“OLED TV\n");
      break ;
}
	#define LCD 1
	#define OLED 2
	
	switch (code) {
	  case LCD:
	      printf ("LCD TV\n");
	      break ;
	  case OLED:
	      printf (“OLED TV\n");
	      break ;
	}
		enum tvtype { LCD, OLED };
		
		enum tvtype code;
		code = LCD;
		
		switch (code) {
		  case LCD:
		      printf ("LCD TV\n");
		      break ;
		  case PDP:
		      printf (“OLED TV\n");
		      break ;
		}

Computers are easy to understand, but people have difficulty remembering .	When writing symbolic constants, easy to make duplication errors.	The compiler checks to ensure that no duplication occurs .

Check points

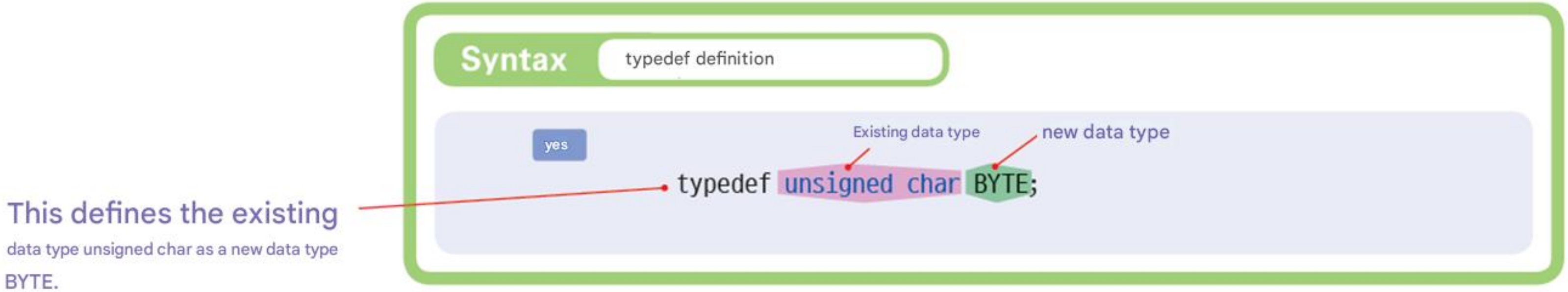
1. The keyword used to declare an enumeration is ______ .
2. In what cases are enumerations used ?
3. If a value is not specifically specified in an enumeration,
   is an integer value automatically assigned ?

The concept of typedef



typedef

`typedef` creates a new name (an alias) for an existing type.
It does *not* create a new type—just a new name for convenience.



Typedef with structure

[Without typedef]

```
struct Person {       // This defines a struct type named Person
    char name[50];
    int age;
};

struct Person s1 =  {"Nora", 27};;  // Now create a variable
```

[Without typedef]

```
struct {                        // This defines an anonymous struct (a struct with no name)
    char name[50];
    int age;
} student;                      // It immediately declares a single variable named student of that anonymous type.
```

```
strcpy (student.name, "Nora");
student.age =  27;
```

~~student s1;~~ // ERROR

```
struct {
    char name[50];
    int age;
} student = {"Nora", 27};
```

Typedef with structure

[With typedef - Recommended style]

```
typedef struct {        // no name (anonymous)
    char name[50];
    int age;
} Person;               // Not variable, now create new struct name (alias)

Person student = {"Nora", 27};

Person student;
Person = (Person){"a", 27};
```

[With typedef]

```
struct Person{        // struct name
    char name[50];
    int age;
};

struct Person student;  // OK

typedef struct Person Student;   // Student becomes a typedef alias for struct Person
Student s = {"Nora", 27};
```

Example of typedef

```
typedef   unsigned char BYTE;
BYTE index; // Same as unsigned int index;

typedef int INT32;
typedef unsigned int UINT32;

INT32 i; // Same as int
UINT32 k; // Same as unsigned int k ;
```

Defining a new type as a struct

- As a structure You can define new types .

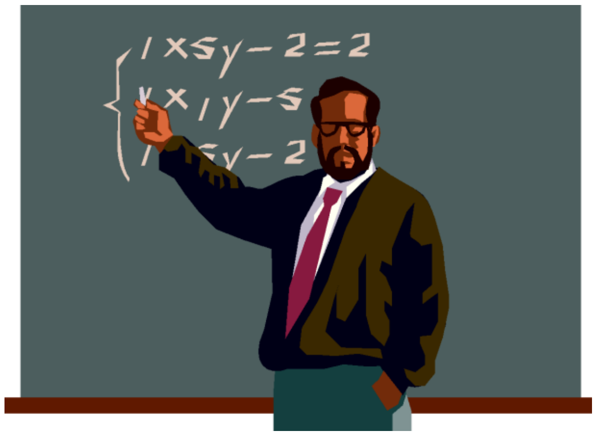
```
struct point {
    int x;
    int y;
};
typedef struct point POINT ;
POINT a, b;
```

```
typedef struct complex {
    double real;
    double image;
} COMPLEX;
COMPLEX x, y;
```

Check points

1. What is the use of typedef ?
2. What are the advantages of typedef ?
3. Let 's define a structure representing an employee and define it as a new type called employee using typedef .

Q & A



char padding (struct)

Size of Struct
What is most efficient way?
Debugger

- Why does a char need 3 bytes of padding even though the smallest memory unit is 1 byte?

  \- Padding exists because of **CPU alignment rules**, not because of byte-accessible memory.

- Why CPUs enforce alignment?

  \- Faster access : Aligned loads/stores are much **faster**.
  \- Hardware requirements : Some CPU architectures **require** aligned access for certain data types.